Detecting Hallucinations in Large Language Models Using Internal Activations and Knowledge-Grounded Reasoning

Pranay Obla **Tavion Fernandes** **Ojas Golatkar** **Kaustubh Mhatre** **Anchalaa Jha**
oblaanan@usc.edu tavional@usc.edu golatkar@usc.edu kmhatre@usc.edu anchalaa@usc.edu

Abstract

Large language models remain susceptible to hallucinations, wherein models generate confident but factually incorrect content. We investigate a layered approach to hallucination mitigation using two open-source instruction-tuned models, Qwen and Gemma, evaluated on the HaluEval benchmark across question answering, dialogue, and summarization tasks. We first fine-tune both models via Low-Rank Adaptation (LoRA) on the TruthfulQA dataset, then apply Self-Consistency decoding to promote grounded outputs, and finally introduce a zero-shot Generator-Checker (G-C) framework in which one model generates responses while the other independently detects and corrects hallucinations. Our results demonstrate that LoRA fine-tuning improves dialogue detection accuracy, and that Self-Consistency yields meaningful F1 gains in summarization. The G-C framework reveals that correction efficacy varies considerably across task types and model pairings, with the Gemma-to-Qwen configuration achieving notably stronger performance than its reverse. These findings highlight model pairing and task type as critical determinants of zero-shot hallucination correction.

1 Introduction

The rapid rise of large language models (LLMs) such as GPT, PaLM, and LLaMA has transformed natural language processing (NLP), enabling strong performance across various tasks (Minaee et al., 2025). However, their widespread deployment has exposed a critical limitation: the tendency to hallucinate, producing responses that are linguistically fluent yet factually incorrect or unsupported by context (McKenna et al., 2023). This phenomenon poses a significant barrier to reliable deployment in high-stakes domains such as healthcare, legal reasoning, and education, where factual accuracy is crucial.

Prior work has addressed hallucination through retrieval-augmented generation (RAG) (Lewis et al., 2020), analysis of internal model activations (Geva et al., 2021), and incorporation of external knowledge bases (Petroni et al., 2019). While effective in constrained settings, these approaches often depend on access to the model's internal components or require substantial architectural changes, which limits their use with off-the-shelf instruction-tuned models. In addition, many existing methods are tested on a narrow range of tasks, leaving it unclear how well these mitigation strategies generalize to structurally different tasks such as question answering, dialogue, and summarization.

We adopt an LLM-as-a-Judge framework, treating hallucination as a binary classification problem across two open-source models, Qwen2.5-3B-Instruct (Qwen et al., 2025) and Gemma-3-4B-IT (Team et al., 2024), evaluated on the HaluEval (Li et al., 2023) benchmark across question answering, dialogue, and summarization tasks. Both models are fine-tuned using Low-Rank Adaptation (LoRA) (Pratap et al., 2025) on the TruthfulQA (Lin et al., 2022) dataset, with only the query and value projection matrices updated, representing approximately 0.12% of total parameters for Qwen and 0.20% for Gemma. Self-Consistency (Wang et al., 2023) decoding is then applied at inference time, generating three candidate outputs per sample and aggregating them via majority voting. A zero-shot Generator-Checker framework is investigated as an independent evaluation of whether LLMs can autonomously detect and correct hallucinations without fine-tuning or supervised signal, wherein one model generates an initial response and a second model judges and provides targeted feedback for refinement.

Our results show that LoRA fine-tuning improves factual grounding but can reduce recall in certain task configurations, while Self-Consistency meaningfully recovers performance, particularly in summarization. The Generator-Checker framework demonstrates that correction efficacy is highly sensitive to model pairing and task type, with the Gemma-to-Qwen configuration substantially outperforming the reverse across all tasks. Collectively, these findings suggest that combining parameter-efficient fine-tuning with inference-time consistency strategies and inter-model verification offers a practical path toward more reliable hallucination detection in open-source LLMs.

2 Related Work

Research on hallucinations in LLMs deal with evaluation and detection. Our use of structured benchmarks is motivated by (McKenna et al., 2023), who show that models hallucinate primarily by taking advantage of shallow lexical shortcuts rather than true reasoning. Their work is merely diagnostic, however we build on it to build an active correction pipeline. The two datasets that are essential to our work are HaluEval (Li et al., 2023) for evaluation and TruthfulQA (Lin et al., 2022) for fine-tuning. Both datasets frame hallucination as a binary classification problem, but we go a level further by combining them in a cross-dataset train-then-evaluate setup that tests whether grounding learned on one distribution transfers to another.

(Snyder et al., 2024) analyze signals such as self-attention scores, hidden-layer activations and softmax distributions during the generation process, rather than evaluating only the generated text . It was found that hallucinated responses exhibit distinct internal activation patterns compared to factual responses and these differences appear from the very first generated token. This means the model often “knows” it is likely to hallucinate before the incorrect content is produced. Self-attention and hidden-layer activations were especially effective for detecting these patterns. However, it focuses on simple factual QA tasks, leaving complex reasoning and longer outputs less explored.

(Kong et al., 2026) introduce HaluGNN, which models each question-answer pair as a weighted directed graph that captures both token level hidden representations and attention dependencies. A GNN is trained to classify between factual and hallucinogenic responses. Even after utilizing minimal labeled data this method achieves competitive performance and also provides token-level interpretability, enabling fine-grained analysis of hallucinated content.

(Petroni et al., 2019) and (Geva et al., 2021) establish that transformers implicitly encode factual knowledge in their parameters but fail to surface it consistently, directly motivating our LoRA fine-tuning approach (Pratap et al., 2025) which selectively updates a small subset of parameters to promote more grounded outputs in Qwen2.5-3B (Qwen et al., 2025) and Gemma-3-4B (Team et al., 2024) without the need for exhaustive retraining. (Lewis et al., 2020) propose Retrieval-Augmented Generation as an alternative grounding mechanism. This technique relies on parametric information that is reinforced by fine-tuning and verified at inference time, instead of using an external retrieval index.

Building on activation based analysis, (Su et al., 2024) propose MIND, an unsupervised framework that leverages contextualized embeddings during generation for real-time hallucination detection. MIND automatically generates training data by prompting models to continue truncation Wikipedia passages and labeling outputs based on entity consistency. It shows that the final-layer embedding of the last generated token is sufficient to classify hallucinations using a lightweight MLP. However, the framework requires model-specific training data and access to hidden representations.

In contrast to internal-state monitoring during generation, (Yadav and Verma, 2026) propose a decoupled architecture for hallucination detection. Instead of analyzing activations during generation, they use frozen encoders like BERT, RoBERTa and DeBERTa to extract embeddings from the final text, which are then classified using lightweight models such as RNNs, LSTMs and GRUs. Their results show that encoder-decoder models can match or outperform much larger generative models in zero-shot hallucinations detection, while using far fewer computational resources. But, the approach works only at the output level and does not analyze generation dynamics, limiting its ability to detect hallucinations during text generation.

The method proposed by (Liu et al., 2025) utilizes a QA based evaluation strategy, which extracts factual statements from the source document and the corresponding pairs of question answers are generated to evaluate the consistency of summaries. FactCC is used to compare answers derived from the source and summary to quantify hallucinations. Finally an iterative refinement mechanism is applied to revise the summary using corrective prompts when inconsistencies are detected. Improved factual consistency and overall summary quality are demonstrated by experimental results on CNN/Daily Mail, PubMed, and ArXiv datasets.

(Wang et al., 2023) propose Self-Consistency as a majority-vote decoding strategy, which we modify diagnostically. Instead of just choosing the best sample, we treat output divergence as a hallucination risk signal that initiates our Generator-Checker correction loop, a combination never before investigated.

3 Problem Description

Large Language Models sometimes produce outputs that are fluent yet factually incorrect. Most existing approaches focus solely on detection, leaving a critical gap which is identifying a hallucinated response does not produce a correct one. Just marking a response as hallucinated or not is insufficient in many cases, as it does not provide a mechanism to improve or validate the output.

Additionally, it is difficult to design a unified framework that detects and corrects hallucinated responses generated by LLM's since hallucination manifests differently across tasks like question answering, dialogue, and summarization. This highlights the need for more robust methods to comprehend and manage hallucinations in LLM generated text.

4 Dataset Description

TruthfulQA (Lin et al., 2022): TruthfulQA is a benchmark designed to measure whether language models generate truthful responses, consisting of questions spanning domains such as health, law, finance, and common misconceptions. For fine-tuning purposes, the dataset was preprocessed by extracting each question alongside its best answer and all additional correct answers, treating each question-answer pair as a separate training example labeled as correct. This expansion from the original question-level format to individual answer-level examples resulted in a total of 8514 training samples. Only correct answers were retained, as the objective was to encourage the model toward factually grounded reasoning rather than expose it to incorrect completions during training.

HaluEval (Li et al., 2023): HaluEval is a large-scale hallucination evaluation benchmark containing human-annotated examples of both hallucinated and factually consistent responses across question answering, dialogue, and summarization tasks, with 10000 samples per task. Each sample contains a context, a correct response, and a hallucinated response. For evaluation, samples were constructed by randomly assigning either the hallucinated or correct response to each example, with the ground truth label set accordingly, producing a balanced evaluation set across all three tasks. No further preprocessing was applied beyond converting the original JSON files to Excel format for compatibility with the data loading pipeline.

5 Methodology

5.1 Baseline Benchmarking

The baseline measured hallucination detection performance of each model without any fine-tuning. For each evaluation sample, we constructed a prompt by concatenating a task-specific instruction with the input context and the candidate response, ending with the token #Your Judgement#. The model generated a greedy response with a maximum of 8 new tokens. The output was parsed by checking for the presence of "Yes" or "No" after stripping punctuation. If both or neither token appeared, the sample was marked as failed. This procedure was applied to both Qwen2.5-3B-Instruct and Gemma-3-4B-IT across all three HaluEval tasks.

5.2 Parameter-Efficient Fine-Tuning using LoRA

We fine-tuned both models using Low-Rank Adaptation (LoRA), which inserts trainable low-rank matrices into selected attention layers while keeping the rest of the model frozen. Adapters were applied to the query and value projection matrices with rank 16, a scaling factor of 32, and a dropout rate of 0.05. Each training example was formatted using the target model's chat template with a system prompt instructing the model to answer accurately and honestly, and sequences were truncated to 512 tokens. Both models were trained for 3 epochs with an effective batch size of 32 (per-device batch size of 4 with 8 gradient accumulation steps), a learning rate of 2e-4, a cosine schedule, and bfloat16 precision. After training, the LoRA weights were merged into the base model for inference and evaluated on the HaluEval benchmark using the same prompt construction and parsing procedure as the baseline.

5.3 Self-Consistency Generation

Self-consistency improves prediction reliability by generating multiple independent outputs for the same input and taking a majority vote. For each evaluation sample, we used the fine-tuned models to generate 3 samples per prompt using stochastic decoding with temperature 0.8, top-p 0.95, and a repetition penalty of 1.1, with a maximum of 128 new tokens. All 3 samples were generated in a single batched forward pass. Each sample was parsed individually for a "Yes" or "No" judgment, and the final prediction was the majority vote across the 3 outputs. We additionally computed lexical and semantic divergence across samples to quantify model uncertainty. Lexical divergence was defined as one minus the mean pairwise token-level F1 score between all sample pairs. Semantic divergence was defined as one minus the mean pairwise cosine similarity between sentence embeddings produced by the all-MiniLM-L6-v2 model. A combined score with equal weight on each metric was recorded alongside the prediction, and samples exceeding a combined divergence threshold of 0.35 were flagged as uncertain, though this did not alter the final judgment.

5.4 Generator-Checker Framework

The generator-checker framework uses two models in a closed-loop refinement process. We used the baseline versions of both models and evaluated both cross-model combinations: Qwen2.5-3B as generator with Gemma-3-4B as checker, and Gemma-3-4B as generator with Qwen2.5-3B as checker. Same-model pairing was deliberately avoided, as a model that confidently produces hallucinations as a generator is unlikely to reliably detect them as a checker, given that both roles would share the same internal biases and knowledge gaps. For each evaluation sample, the checker was given the task-specific instruction, the input context, and the candidate response, and was instructed to output either "Yes" or "No" followed by a "Feedback" line describing which specific claims were unsupported, if applicable. The checker generated outputs greedily with a maximum of 128 new tokens. If the checker returned "No," the loop terminated and that verdict was recorded as the final judgment. If the checker returned "Yes," the feedback was passed to the generator along with the original context and the flagged response, prompting it to produce a revised response. The checker then evaluated the revised response and the loop terminated regardless of the outcome, allowing at most one refinement round per sample and at most two checker calls total. Generator outputs used temperature 0.7 and top-p 0.9 with a maximum of 256 new tokens.

5.5 Evaluation Metrics

All experiments in the detection pipeline, covering the base, LoRA, and SC-LoRA configurations, are evaluated using accuracy and F1 score. Accuracy measures the overall proportion of correctly classified samples, while F1 score is computed with respect to the positive class, where Yes indicates a hallucinated response. Given the binary nature of the task and the potential for class imbalance, F1 is treated as the primary metric as it more faithfully reflects a model's ability to identify hallucinated responses rather than simply predicting the majority class.

For the Generator-Checker framework, detection rate and correction rate are used instead. Since the framework does not operate as a standard classifier but rather as an iterative refinement pipeline, accuracy and F1 are not well suited to capturing its behavior. Detection rate measures the proportion of truly hallucinated samples that the checker correctly identifies in the first round, while correction rate measures the proportion of detected hallucinations that are subsequently revised into a non-hallucinated response after feedback.

6 Experimental Results

6.1 Hallucination Detection across Base, LoRA, and Self-Consistency Configurations

Tables 1, 2, and 3 report accuracy and F1 scores across all tasks and methods.

Question Answering: All configurations perform near chance level (0.50) in accuracy, suggesting the task is particularly challenging under binary classification. Gemma's base model achieves a notably stronger F1, which collapses substantially after LoRA fine-tuning before partially recovering under SC-LoRA. Qwen shows a more stable progressive improvement across configurations, though absolute performance remains low throughout.

Dialogue: Both models perform most reliably on this task at baseline. Gemma's base model achieves the strongest single result across all tasks in both accuracy and F1, however LoRA again causes a

Model	Configuration	Accuracy	F1
Qwen	Base	0.4962	0.0217
	LoRA	0.5022	0.0812
	SC-LoRA	0.5026	0.1483
Gemma	Base	0.5012	0.6367
	LoRA	0.5012	0.0739
	SC-LoRA	0.4943	0.3152

Table 1: Results on the QA task

severe collapse in F1. Qwen remains comparatively stable across configurations, with SC-LoRA yielding the strongest Qwen result on this task.

Model	Configuration	Accuracy	F1
Qwen	Base	0.5869	0.4234
	LoRA	0.5870	0.3510
	SC-LoRA	0.5892	0.4186
Gemma	Base	0.6125	0.6597
	LoRA	0.5204	0.0968
	SC-LoRA	0.5333	0.3662

Table 2: Results on the dialogue task

Summarization: This task shows the most pronounced impact of Self-Consistency. LoRA collapses F1 to near zero for both models, while SC-LoRA substantially recovers both accuracy and F1 across both models. Summarization thus represents the task where Self-Consistency provides the greatest absolute gain, suggesting it is particularly well suited to the longer and more complex reasoning this task requires.

Model	Configuration	Accuracy	F1
Qwen	Base	0.2975	0.4565
	LoRA	0.3629	0.0315
	SC-LoRA	0.4610	0.3662
Gemma	Base	0.3869	0.6200
	LoRA	0.3625	0.0000
	SC-LoRA	0.4876	0.3841

Table 3: Results on the summarization task

6.2 Hallucination Detection and Correction using Generator-Checker Framework

Table 4 presents detection and correction rates for each model pairing across tasks.

Gemma-to-Qwen: This pairing demonstrates meaningful detection and correction across all tasks. In QA, Qwen as checker detects the majority of hallucinations, though correction remains limited. In dialogue, detection is considerably lower, yet when hallucinations are detected, correction is exceptionally high, suggesting the generator responds well to checker feedback in conversational contexts. Summarization yields a balanced outcome, with both detection and correction performing reasonably well.

Qwen-to-Gemma: This pairing performs near zero across all tasks in both detection and correction, with the sole partial exception of dialogue correction. Gemma as checker appears unable to reliably flag hallucinations in Qwen's outputs under zero-shot conditions, and consequently Qwen receives little actionable feedback for refinement.

Generator → Checker	Task	Detection Rate	Correction Rate
Gemma → Qwen	QA	0.7759	0.3150
	Dialogue	0.1950	0.9104
	Summarization	0.5112	0.7400
Qwen → Gemma	QA	0.0000	0.0000
	Dialogue	0.0078	0.1250
	Summarization	0.0019	0.0000

Table 4: Results across all model pairings and tasks

7 Conclusion

This work investigated a layered approach to hallucination detection in open-source LLMs, combining LoRA fine-tuning, Self-Consistency decoding, and a zero-shot Generator-Checker framework across question answering, dialogue, and summarization tasks on HaluEval. Our results demonstrate that while LoRA fine-tuning consistently disrupts base model detection behavior, Self-Consistency serves as an effective recovery mechanism, particularly in summarization. The Generator-Checker framework reveals that inter-model verification is highly asymmetric, with model pairing emerging as a critical and non-trivial design choice.

Limitations: This work operates within several practical constraints. Fine-tuning was conducted on a single dataset that differs in structure from the evaluation benchmark, which likely introduces some distribution mismatch. The Generator-Checker framework was evaluated under zero-shot conditions with a single refinement round, and experiments were confined to two relatively small models, limiting the breadth of conclusions that can be drawn.

Future Work: Exploring a broader range of fine-tuning datasets and task formats presents a promising direction for improving detection stability across configurations. More sophisticated Self-Consistency aggregation strategies, such as confidence-weighted voting or semantic clustering, could further strengthen reliability. Extending the Generator-Checker framework to multiple rounds of iterative refinement and incorporating fine-tuned checker models rather than zero-shot prompting are natural next steps toward stronger correction performance. Scaling these experiments to larger models would additionally help clarify whether the asymmetries observed between model pairings reflect fundamental model differences or are influenced by relative model scale.

Individual Contributions

Pranay Obla performed LoRA fine-tuning of both Qwen and Gemma, tested the fine-tuned models on HaluEval, and designed and implemented the full Generator-Checker framework including its evaluation script.

Tavion Fernandes conducted benchmarking of both Qwen and Gemma on HaluEval and contributed to the Generator-Checker evaluation script.

Ojas Golatkar developed the evaluation script for benchmarking and jointly developed the Self-Consistency decoding pipeline and conducted the corresponding evaluations.

Kaustubh Mhatre developed the evaluation script for the LoRA fine-tuned models and jointly developed the Self-Consistency decoding pipeline and conducted the corresponding evaluations.

Anchalaa Jha was responsible for dataset preparation, including preprocessing TruthfulQA and HaluEval and performing the necessary data transformations for model compatibility, and jointly developed the Self-Consistency decoding pipeline and conducted the corresponding evaluations.

All authors contributed equally to the documentation and preparation of this report.

References

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2021. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Linggang Kong, Yunlong Zhang, Xiaofeng Zhong, Haoran Fu, Yongjie Wang, and Huijun Liu. 2026. Halugnn: Hallucination detection in large language models using graph neural network. *Expert Systems with Applications*, 306:130857.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.

Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. 2023. Halueval: A large-scale hallucination evaluation benchmark for large language models. *Preprint*, arXiv:2305.11747.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Truthfulqa: Measuring how models mimic human falsehoods. *Preprint*, arXiv:2109.07958.

Shenling Liu, Yang Gao, ShaSha Li, PanCheng Wang, and Ting Wang. 2025. A hallucination detection and mitigation framework for faithful text summarization using llms. *Scientific Reports*, 16(1):1374.

Nick McKenna, Tianyi Li, Liang Cheng, Mohammad Hosseini, Mark Johnson, and Mark Steedman. 2023. Sources of hallucination by large language models on inference tasks. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 2758–2774, Singapore. Association for Computational Linguistics.

Shervin Minaee, Tomas Mikolov, Narjes Nikzad, Meysam Chenaghlu, Richard Socher, Xavier Amatriain, and Jianfeng Gao. 2025. Large language models: A survey. *Preprint*, arXiv:2402.06196.

Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H. Miller, and Sebastian Riedel. 2019. Language models as knowledge bases? *CoRR*, abs/1909.01066.

Samar Pratap, Alston Richard Aranha, Divyanshu Kumar, Gautam Malhotra, Anantharaman Palacode Narayana Iyer, and Shylaja S.S. 2025. The fine art of fine-tuning: A structured review of advanced llm fine-tuning techniques. *Natural Language Processing Journal*, 11:100144.

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, and 25 others. 2025. Qwen2.5 technical report. *Preprint*, arXiv:2412.15115.

Ben Snyder, Marius Moisescu, and Muhammad Bilal Zafar. 2024. On early detection of hallucinations in factual question answering. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '24, page 2721–2732. ACM.

Weihang Su, Changyue Wang, Qingyao Ai, Yiran HU, Zhijing Wu, Yujia Zhou, and Yiqun Liu. 2024. Unsupervised real-time hallucination detection based on the internal states of large language models. *Preprint*, arXiv:2403.06448.

Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, Pouya Tafti, Léonard Hussenot, Pier Giuseppe Sessa, Aakanksha Chowdhery, Adam Roberts, Aditya Barua, Alex Botev, Alex Castro-Ros, Ambrose Slone, and 89 others. 2024. Gemma: Open models based on gemini research and technology. *Preprint*, arXiv:2403.08295.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. *Preprint*, arXiv:2203.11171.

Sourabh Yadav and Nishchal Verma. 2026. A hybrid framework for hallucination detection in large language models. *IEEE Transactions on Artificial Intelligence*, PP:1–13.